Wydział Mechaniczny Energetyki i Lotnictwa

The Impact of Hydrogen Addition on Laminar Flame Speed and NO_x Emissions in Methane-Air Mixtures

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June 15, 2026

Abstract

Hydrogen-enriched methane combustion has attracted significant attention due to its potential for improving combustion efficiency and reducing carbon dioxide emissions. However, the addition of hydrogen also affects flame propagation and pollutant formation. In this work, numerical simulations were performed using the Cantera library and the GRI-Mech 3.0 chemical kinetic mechanism to investigate the influence of hydrogen content on laminar flame speed and nitrogen oxides emissions in methane-air mixtures. The effects of equivalence ratio, pressure and initial temperature were also considered. The obtained results provide insight into the advantages and limitations of hydrogen blending in natural gas combustion systems.

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1 Introduction

Natural gas is one of the most widely used fuels in domestic and industrial applications. Methane, which is its primary component, produces lower carbon dioxide emissions compared to coal and oil. Nevertheless, further reduction of greenhouse gas emissions is required to meet environmental regulations and climate targets.

Hydrogen has emerged as a promising additive to methane because it does not contain carbon and exhibits high reactivity. Introducing hydrogen into methane-air mixtures can enhance combustion characteristics, increase flame stability and improve combustion efficiency. On the other hand, higher flame temperatures may promote the formation of nitrogen oxides $NO(_x)$, which are harmful pollutants.

The objective of this study is to investigate the influence of hydrogen addition on laminar flame speed and $NO(_x)$ formation in methane-air mixtures using detailed chemical kinetics.

2 Literature Review

Hydrogen-enriched combustion has been extensively investigated in recent years due to its relevance for future low-carbon energy systems.

According to Turns (2012), hydrogen addition significantly increases flame speed because of its high diffusivity and rapid reaction kinetics. Similar observations were reported by Law (2006), who showed that hydrogen-containing fuels exhibit enhanced flame propagation characteristics.

Numerical investigations using the GRI-Mech 3.0 mechanism have demonstrated that increasing hydrogen concentration generally results in higher adiabatic flame temperatures and consequently increased thermal NO formation. Studies by Konnov (2008) and others indicated that the amount of NO produced strongly depends on equivalence ratio and combustion temperature.

Detailed chemical kinetic mechanisms implemented in Cantera have become a standard approach for predicting laminar flame properties and pollutant emissions.

3 Model Description

The simulations were performed using Python and the Cantera software package. The GRI-Mech 3.0 reaction mechanism was employed to describe methane-hydrogen combustion chemistry.

The initial conditions were:

- Temperature: 300 K
- Pressure: 1 atm
- Equivalence ratio: $\phi = 1.0$

Hydrogen concentration in the fuel mixture was varied from 0% to 50%. The numerical model solves conservation equations together with detailed chemical kinetics in order to determine the one-dimensional freely propagating flame.
The quantities analyzed in this study include:

- Laminar flame speed;
- Adiabatic flame temperature;
- NO concentration;
- $NO(_2)$ concentration;
- $N(_2)O$ concentration.

4 Results and Discussion

4.1 Influence of Hydrogen Fraction

Hydrogen addition increases the laminar flame speed due to the higher reactivity and diffusivity of hydrogen compared with methane. Consequently, combustion becomes faster and more stable.

Figure 1 presents the variation of laminar flame speed as a function of hydrogen content.

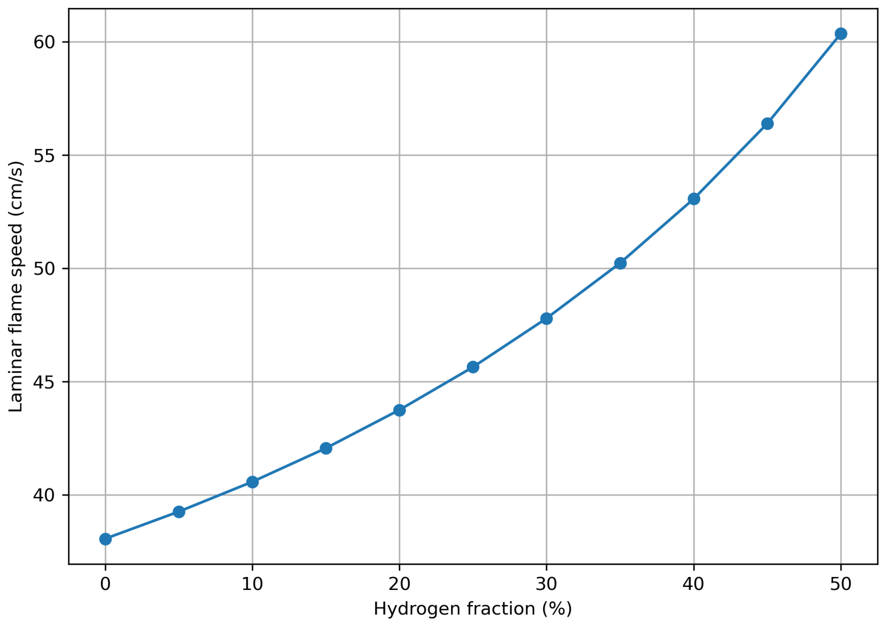

Figure 1: Laminar flame speed versus hydrogen fraction.

A nearly monotonic increase in flame speed is expected with increasing hydrogen concentration.

4.2 Adiabatic Flame Temperature

Hydrogen enrichment increases the adiabatic flame temperature due to higher reaction rates and more complete combustion.

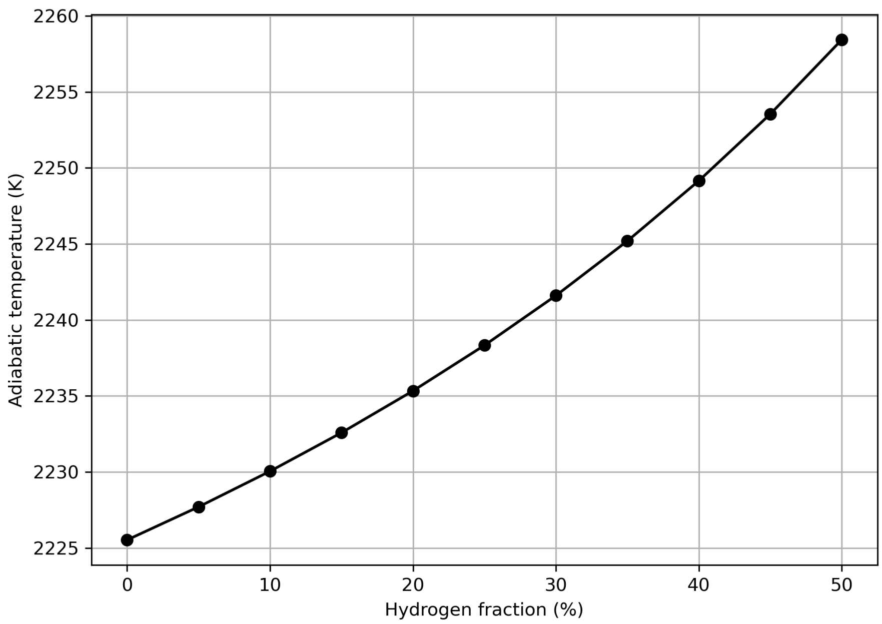

Figure 2: Adiabatic flame temperature versus hydrogen fraction.

Higher flame temperatures have a direct impact on pollutant formation, especially NO_x species.

4.3 NO Emissions

The increase in combustion temperature promotes thermal NO formation through the Zeldovich mechanism.

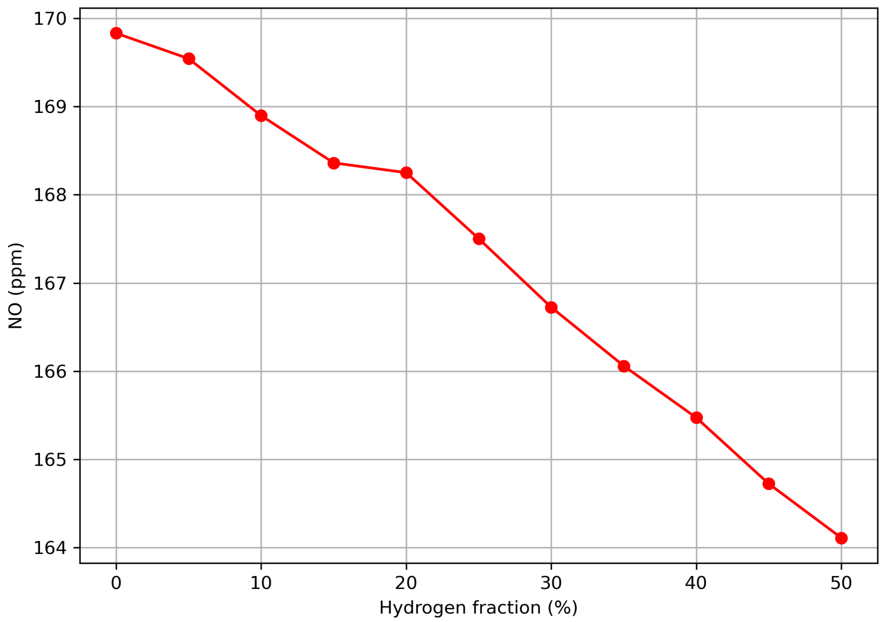

Figure 3: NO concentration as a function of hydrogen fraction.

4.4 NO_2 Emissions

NO_2 formation is strongly dependent on post-flame oxidation of NO and local temperature fields.

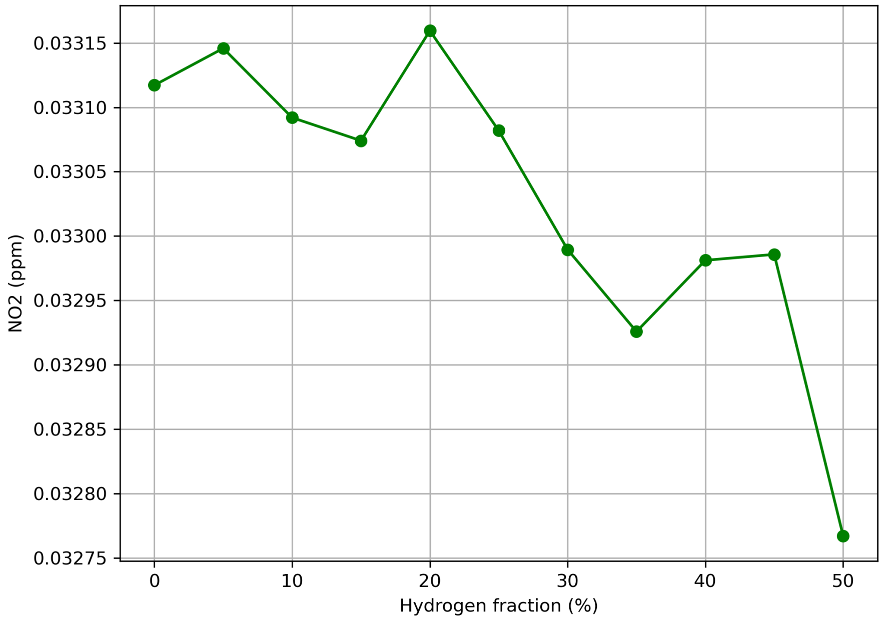

Figure 4: NO_2 concentration as a function of hydrogen fraction.

4.5 N_2O Emissions

Nitrous oxide (N_2O) is typically formed in smaller quantities but remains an important greenhouse gas.

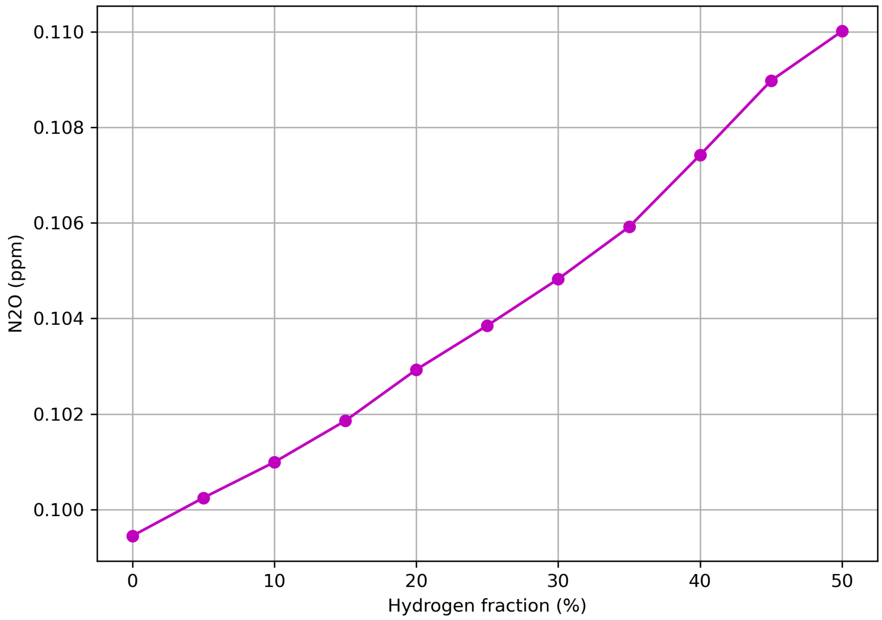

Figure 5: N_2O concentration as a function of hydrogen fraction.

4.6 Summary of Trends

Across all cases, hydrogen addition increases flame speed and adiabatic flame temperature. While combustion performance improves, the emission trends of pollutant NO_x species show conflicting behaviors, with NO decreasing and N_2O increasing.

5 Conclusions

Numerical simulations were carried out to study the effect of hydrogen addition on methane-air combustion. The following conclusions can be drawn:

1. Hydrogen addition significantly increases the laminar flame speed, rising from approximately 38 cm/s at 0% H_2 to over 60 cm/s at 50% H_2.
2. Higher hydrogen fractions result in increased adiabatic flame temperatures, elevating the temperature from 2225 K to over 2258 K.
3. In contrast to standard thermal NO_x theory (Zeldovich mechanism), the simulations show a distinct decrease in NO concentration from around 169.8 ppm to 164.1 ppm at 50% hydrogen. This indicates that chemical dilution or reduced residence time outweighed the thermal effect.
4. NO_2 emissions exhibit a non-linear, overall decreasing trend at higher hydrogen fractions, remaining within a very low range (0.03275–0.03316 ppm).
5. N_2O emissions show a nearly linear and steady increase with hydrogen addition, rising from 0.099 ppm to 0.110 ppm.
6. Hydrogen-enriched methane combustion offers clear advantages for future low-carbon energy systems by reducing carbon emissions, while the trade-offs in specific NO_x species (N_2O) remain an important operational challenge.

References

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